

ELEC5305 Project Proposal

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1 Project Title

Performance Evaluation of VAD-based Wiener Filtering for Speech Enhancement in Stationary vs. Non-stationary Noise Environments.

2 Student Information

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3 Project Overview

This project aims to develop, simulate and evaluate a Wiener filter voice enhancement system based on voice activity detection (VAD), which is implemented in MATLAB. The main research problem is to quantitatively analyze the fluctuations of the effectiveness of Wiener filters under different types of noise: specifically, steady-state noise (such as constant white noise generated by air conditioners) and non-steady-state noise (such as fluctuating street background noise or noise in crowded areas).

Improving voice recognition in noisy environments is still a key challenge in the fields of telecommunications, hearing aid design and voice control human-computer interface. This project aims to propose an efficient and lightweight MATLAB solution by isolating and evaluating the performance of algorithms in different noise environments, while better understanding the basic limitations of classical statistical signal processing methods in dynamic acoustic environments.

4 Background and Motivation

The voice recorded in the real environment will inevitably be interfered by various background noises, which seriously reduces the quality and understandability of the voice. Historically, classic signal processing technology has been used to solve such problems. Among them, Wiener filter is one of the most prominent methods. The principle of this filter is: running in the frequency domain, the gain function is applied to the noise speech audio spectrum to minimize the mean square error (MSE) between the estimated voice signal and the pure voice signal.

A key prerequisite for the effectiveness of the Wiener filter is whether the noise power spectrum density (PSD) can be accurately estimated. In practical applications, this is usually achieved by using a voice activity detector (VAD). VAD recognizes frames without voice (frames with noise only) and updates the noise characteristics accordingly. Although this method is very effective for steady-state noise (its statistical characteristics do not change over time) and has high computational efficiency, its performance usually decreases significantly in non-steady-state noise environments.

The latest progress in the field of deep learning has introduced a highly robust noise suppression model. However, these models often require a lot of computing resources and are not always suitable

for low-power embedded systems. Therefore, it is of great practical significance to optimize classic algorithms such as Wiener filters. I chose this topic because it can combine basic auditory signal processing theory with actual engineering evaluation, so that it can deeply explore short-time Fourier transform (STFT) and statistical masking technology, and realize it with the help of the core MATLAB toolbox.

5 Proposed Methodology

The project will be operated in MATLAB, using the core coding modules provided by the ELEC5305 course, especially based on Week4-STFTExample.mlx and Week6C-VAD and WienerMask.mlx .

1.Data preparation: The data source will contain cleaned voice fragments selected from standard open source corpus (such as Noizus or Timit Corpus). These cleaned signals will be interfered by human ways to mix with specific noise characteristics, and the signal-to-noise ratio (SNR) will vary, such as 0 dB, 5 dB and 10 dB. Noise will be divided into the following categories:

Fixed noise: Gaussian white noise, fan noise.

Non-steady noise: street traffic noise, noise in crowded areas.

2.Signal processing process:

Time-frequency transformation: Use short-time Fourier transform (STFT) to convert noisy time-domain signals into time-frequency domains. Appropriate window functions (such as Hamming windows) and overlap-plus (OLA) techniques will be configured to ensure perfect reconstruction.

Voice Activity Detection (VAD): An energy-based VAD algorithm will be adopted to classify frames into two types: with or without voice.

Noise estimation and masking: During the voice missing frame, the noise power spectrum density will be updated. Then, a Wiener filter masking will be calculated based on the estimated a priori and post-post-sense signal-to-noise ratio.

Reconstruction: Wiener masking will be applied to the noise-containing value spectrum, and the enhanced voice will be combined with the original noise-containing phase and recalculated to the time domain through the inverse short-time Fourier transform (ISTFT).

6 Expected Outcomes

The final deliverables of this project will include:

1. A practical MATLAB simulation tool: a complete and executable MATLAB script, which includes the whole process from noise mixing to voice enhancement and indicator evaluation.
2. Performance indicators and analysis: a comprehensive evaluation report, including quantitative indicators, mainly the improvement of objective signal-to-noise ratio (SNR) under different noise conditions.
3. Visual comparison: high-resolution spectrum map shows the time-frequency energy distribution of clean signals, noise-containing signals and enhanced signals, providing qualitative visual evidence for the effectiveness of the filter.
4. GitHub repository and presentation: a detailed public GitHub repository containing all source code, required README.md files, final written reports, and video presentation links delivered through GitHub Pages.

7 Timeline (Weeks 1–13)

Week	Task
1-2	Select the project topic, prepare the GitHub repository, and set up the GitHub page.
3-5	Conduct a literature review, prepare a pure voice data set suitable for this project (such as TIMIT), and collect various required noise samples.
6-7	Use the course module (Week4-STFTExample.mlx) to build a core STFT/ISTFT framework and prepare relevant noise mixing scripts. Integrate the voice activity detector (VAD) and sort out the Wiener filter estimation logic (Week6C-VADandWienerMask.mlx).
8-9	Conduct systematic tests in two noisy environments. Calculate the improvement of the signal-to-noise ratio and generate a comparative spectrum chart.
10-11	Complete the code optimization, write the final project report, record the video presentation content, and complete the writing of GitHub documents.
12-13	

8 References

1. Loizou, P. C. (2013). *Speech Enhancement: Theory and Practice*.
2. Reddy, C. et al. (2021). DNS Challenge: Improving Noise Suppression Models.
3. Ephraim, Y., & Malah, D. (1984). Speech enhancement using a minimum mean-square error short-time spectral amplitude estimator.